

PRINCE RAJ

Software Engineer | Full-Stack & AI/ML

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SUMMARY

Software Engineer with 2 years building production front-end and AI/LLM systems. Strong in **React.js**, **Next.js**, and **TypeScript**, with **FastAPI** backend work and hands-on **RAG**, **LangChain**, and **LLM orchestration**. M.Tech CSE, NIT Jalandhar; GATE and UGC NET 2025 qualified.

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

Master of Technology (M.Tech), Computer Science and Engineering | CGPA: 8.46/10

Jalandhar, India

Jul 2024 - Jun 2026

IKG Punjab Technical University (Main Campus)

Bachelor of Technology (B.Tech), Computer Engineering | CGPA: 8.39/10

Kapurthala, India

Jul 2019 - Jul 2023

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL

Frontend: React.js, Next.js, Redux, Zustand, Tailwind CSS, Material UI, Shadcn/UI, HTML5, CSS3

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Microservices

AI/ML & GenAI: LangChain, RAG, LLM Pipelines, Prompt Engineering, OpenAI API, Llama 3, Whisper, Embeddings

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch, Firebase

Cloud & DevOps: AWS, Docker, Vercel, Git/GitHub, CI/CD, Linux

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Testing: Jest, React Testing Library, Postman, Unit Testing, Debugging

Practices: Agile/Scrum, Code Review, Technical Documentation, Cross-functional Collaboration, Communication, Problem Solving

Languages: English (Professional), Hindi (Native)

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Sep 2025 - Feb 2026

Ayudo.ai (AI-Powered Customer Support Platform)

Noida, India

- Led full-stack development of a **4-channel** AI support platform (WhatsApp, Email, Slack, Voice) in **React.js** and **Next.js**.
- Built a **RAG** knowledge base chatbot over internal docs with **LangChain** embeddings, cutting first-response time **40%**.
- Engineered a drag-and-drop workflow builder integrating **2,700+ apps** via agentic AI pipelines with role-based access control.
- Delivered **6 FastAPI** microservices for LLM orchestration, **WebSocket** messaging and SSO/OAuth, containerized with **Docker**.
- Partnered with product and design across **2-week Agile** sprints, reviewing pull requests and documenting API contracts.

Software Engineer

Jul 2023 - Jul 2024

Department of Holistic Education

Mohali, India

- Owned the **React** frontend of a real-time log analytics platform used by **5** engineering teams to debug production issues.
- Architected a **20+ component** design system with **Redux**, interactive charts, a trace explorer, and role-based views.
- Integrated search, filtering, and distributed-trace views into the dashboard UI, cutting engineering debugging time **60%**.
- Shipped server-side-rendered **Next.js** apps for thousands of active users, lifting **Lighthouse** performance scores above **90**.
- Optimized Android applications in **Java** using **MVVM** architecture, reducing load time by **30%** and improving retention.

Graduate Engineer Trainee

Jan 2023 - Jun 2023

Department of Holistic Education

Mohali, India

- Developed **React.js** frontend modules for responsive UI, form validation, and API-driven rendering; promoted within **6 months**.
- Resolved **40+** cross-browser UI defects across Chrome, Firefox, Safari; refactored shared styles into reusable **CSS** modules.
- Streamlined **Agile** sprint planning and technical documentation, raising frontend delivery efficiency **15%**.

PROJECTS

AI Resume Tailor & Interview Prep | 🐙 GitHub | 🎥 Live Demo

Next.js, Llama 3, RAG

- Implemented an ATS-aware resume analyzer with **Llama 3** semantic matching and a **RAG** pipeline over job descriptions.
- Improved resume-to-job alignment scores by **40%** through prompt engineering and automated skill-gap analysis.
- Generated personalized interview questions and career recommendations via **LLM** pipelines.
- Designed the responsive **Next.js** UI with resume upload, side-by-side JD comparison, and live keyword-match highlighting.

Real-Time Collaborative Whiteboard | 🐙 GitHub | 🎥 Live Demo

Next.js, WebSockets, Redis

- Scaled a multi-user canvas to **20+** concurrent editors with state syncing via **WebSocket** clusters and **Redis** Pub/Sub.
- Eliminated edit conflicts with operational transformation, plus live cursor tracking, presence detection and undo/redo.
- Structured the **React** component architecture and canvas rendering layer for smooth concurrent drawing.
- Tuned canvas re-rendering with layered redraws and event batching to hold **60 fps** under high-frequency input.

Real-Time AI Voice Agent | 🐙 GitHub

FastAPI, LangChain, Whisper, Docker

- Launched a conversational AI agent on **WebSockets** and **faster-whisper**, streaming speech-to-text in under **500 ms**.
- Deployed **LangChain** LLM pipelines for audio responses in **Docker** containers supporting **10+** concurrent streams.
- Added streaming text-to-speech playback and barge-in handling for natural mid-response turn-taking.
- Managed conversation state and context windows to keep multi-turn dialogue coherent within token limits.

ACHIEVEMENTS

- GATE 2025 Computer Science** - Qualified | **UGC NET 2025 Computer Science** - Qualified
- Solved **200+** Data Structures and Algorithms problems on LeetCode and GeeksforGeeks.